# Yuanhao Zheng

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Research Interests: Fruits quality detection, Natural products utilization

## EDUCATION

**Zhejiang University (ZJU)** ***Hangzhou, China***

M.Eng. in Agricultural Engineering | Grade: 88.6/100 *09/2022 – 06/2025 (Expected)*

Agricultural Systems Modeling and Big Data Analytics (93), Bio-systems Detection and Control Technologies (93), Engineering Analysis of Biological System (98), Special Topic of Precision Farming Technology (88)

**Zhejiang University (ZJU)** ***Hangzhou, China***

B.Eng. in Food Science & Engineering | GPA: 3.61/4.0 *09/2017 – 06/2021*

Analytical Chemistry (89), Fermentology (95), Food Microbiology (86), Food Sensory Evaluation (94), Fruit and Vegetable Preservation Engineering (89), Food Additives (92), Dairy Technology (93), Meat Technology (90)

## PUBLICATIONS

[1] **Zheng, Y.**, Luo, X., Gao, Y., Sun, Z., Xu, H., & Xie, L.* Lycopene detection in cherry tomatoes considering feature enhancement and data fusion. ***Food Chemistry***, 2024. (Under review)

[2] **Zheng, Y.**, Liu, P., Zheng, Y., & Xie, L.* Improving SSC detection accuracy of cherry tomatoes by feature synergy and complementary spectral bands combination. ***Postharvest Biology and Technology***, 2024. [DOI]

[3] **Zheng, Y.**, Zhou, Y., Liu, P., Zheng, Y., Wei, Z., Li, Z., & Xie, L.* Improving discrimination accuracy of pest-infested crabapples using Vis/NIR spectral morphological features. ***Journal of Food Measurement and Characteristic***, 2024. (Under review)

[4] Li, J., **Zheng, Y.**, Wang, P.*, & Zhang, H.* The alginate dialdehyde crosslinking on curcumin-loaded zein nanofibers for controllable release. ***Food Research International***, 2024. [DOI]

[5] **Patent:** Xie, L., **Zheng, Y.**, Xu, H., Gao, T. Detection equipment and method for carotenoid content in fruits. China Patent Application, 2023116457571. filed Dec 4th, 2023. (Patent Pending)

## RESEARCH EXPERIENCE

**[1] Nondestructive detection of cherry tomatoes' quality** *06/2022 - 08/2024*

- Improved nondestructive prediction performance for lycopene and soluble solids content (SSC) in cherry tomatoes based on UV/Vis/NIR spectroscopy and machine vision.
- Developed feature synergy and spectral bands combination strategy to predict SSC.
- Implemented spectral characteristic bands enhancement and spectrum-image fusion to enhance lycopene content detection accuracy, achieving prediction $R^2$ of 0.95.
- Detected fruit quality from the perspective of supplement and integrity for raw information.

**[2] Pest-infested crabapples discrimination using Vis/NIR spectroscopy.** *10/2023-08/2024*

- Discriminated pest-infested crabapple through compensating for the influence of external orientation and enhancing the difference in spectral morphological features (SMFs).
- Utilized the global model incorporating various postures to mitigate the external impact of crabapple orientation on the classification model.

- Investigated and applied SMFs to amplify the differences between healthy and pest-infested crabapples' spectra, achieving a discrimination accuracy of 95.94%.
- Classified crabapples in terms of influence elimination and difference amplification.

**[3] The electrospun zein nanofibers crosslinked by alginate dialdehyde for controllable curcumin release** (Undergraduate Thesis, Advisor: Prof. Hui Zhang) *10/2020 – 06/2021*

- Prepared zein/alginate dialdehyde (AD) electrospun nanofibers through green crosslinking.
- Enhanced the nanofiber's physical/mechanical properties, hydrophobicity, and swelling ratio, while reduced weight loss in water through high cross-linking degree.
- Loaded curcumin (CUR) with nanofiber, achieved effective encapsulation and controlled release, and noted a gradual antioxidant behavior rise corresponding to the release behavior.
- Constructed electrospun membrane for the protection and controlled release of phytochemicals.

**[4] Extraction and purification of strawberry anthocyanin and preparation of its embedded microemulsion** (Undergraduate Student Research Training) *05/2019 – 09/2020*

- Optimized the ultrasound-assisted extraction of strawberry anthocyanins using response surface methodology, and separated and purified anthocyanins with Macroporous resin and Sephadex gel.
- Developed W/O emulsions with Tween 80 and Span 20 to encapsulate anthocyanins.

## ACADEMIC ACTIVITIES

**Services**

- Part-time counselor for undergraduate students in 1st year *08/2023 – 07/2024*
- Peer review service for ***Food Control*** 05/2024 - present

**Awards**

- National Encouragement Scholarship *2020*
- Scholarship for Excellence in Special Major *2019*
- The Third Prize Scholarship of ZJU *2019*

**Conference Presentations**

- World Photonics Conference (**AOPC2024**) *07/2024*
  Lycopene detection in cherry tomatoes, Oral presentation ***Beijing, China***
- International Commission of Agricultural and Biosystems Engineering (**CIGR 2024**) *05/2024*
  SSC prediction in cherry tomatoes, Oral presentation **Jeju, South Korea**
- Chinese Society of Agricultural Engineering (**CSAE 2023**) *08/2023*
  Vis/NIR-based cherry tomato quality detection, Poster presentation **Chengdu, China**

## SKILLS

**Language:** Mandarin (native), English (fluent)

**Program Skills:** C/C++, MATLAB, Python

**Experimental Skills:** Machine learning, Near-infrared Spectroscopy, Spectral data analysis, Extraction of natural products, Chemical analysis, Image processing.